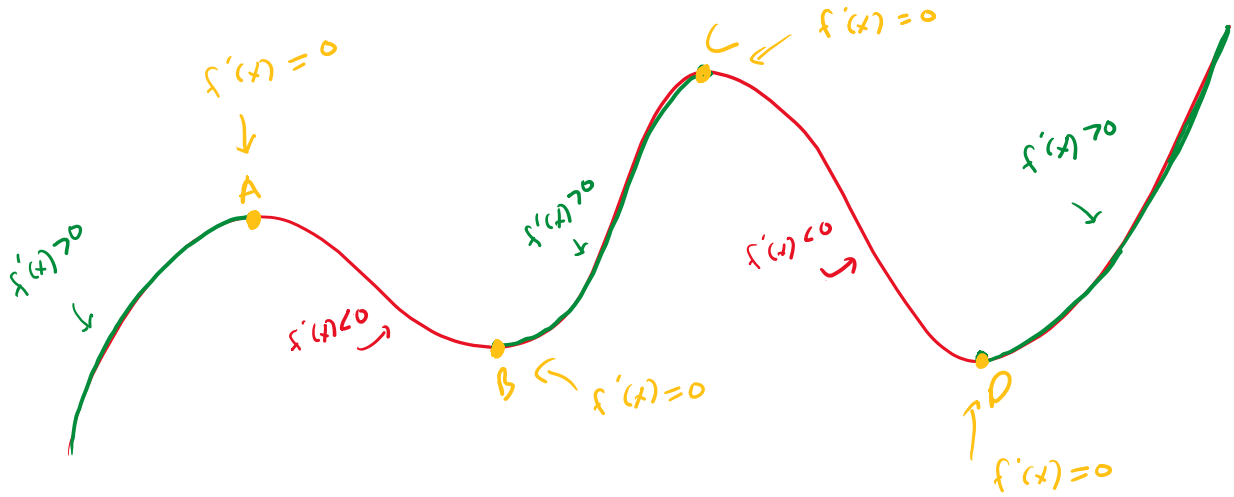
Applications of Derivatives

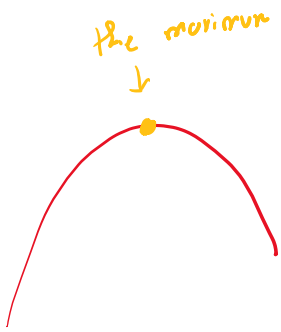


A ; C are local maxima

B ; D are local minima

A, B, C and D are all critical points.

(*) If f(x) has ONLY one critical point:



OR

Finding the maximum or minimum of a function.

We can use the derivatives to find the minimum and maximum of functions. Suppose we want to find the minimum and maximum of a function $f(x)$. In these examples, we will be given that $f(x)$ will be maximized or minimized at the soltion of $f'(x) = 0$. Therefore, the procedure to do is as follows.

- Step 1: Find and simplify $f'(x)$
- Step 2: Solve $f'(x) = 0$. Suppose that the solution is $x = c$. (the only critical point)
- Step 3: Confirm that $\underbrace{f''(c)}_{\text{second derivative}} < 0$ for the maximum and $f''(c) > 0$ for the minimum.

1. **Cost Function, Marginal Cost Function**

   The marginal cost represents the approximate cost of producing the next unit.

Example 1:

The per-day cost function of the manufacture of portable MP3 players is given by

C(x) = 128,000 + 30x + x^3

where x is the number of MP3 players manufactured per day. Assume that the company cannot manufacture more than 100 MP3 players per day.

a. Find the $\underbrace{\text{marginal Cost}}_{MC}$ function.

$$MC = C'(x) = (128{,}000 + 30x + x^3)'$$

$$= (128{,}00)' + (30x)' + (x^3)'$$

$$= 0 + 30 + 3x^2$$

$$\boxed{MC = 3x^2 + 30}$$

b. Interpret the meaning of C' (80) or MC(80)

$MC(80) = 3 * 80^2 + 30 = \boxed{19230}$

$MC(80) \approx$ the cost of making the 81st items.

The marginal cost represents the approximate cost of producing the next unit.

c. Find the actual cost of producing the 81st item.

$C(81)$: cost of making the first 81 items.

$C(80)$: cost of making the first 80 items.

Actual cost of making the 81st item is

$C(81) - C(80) = 661871 - 642400$

$= \boxed{19471}$

$C(x) = 128{,}000 + 30x + x^3$

d. Write the $\underbrace{\text{average cost}}_{AC}$ function

$$AC = \frac{\text{cost of making } x \text{ items}}{x} = \frac{128{,}000 + 30x + x^3}{x}$$

e. How many MP3 players should be manufactured in order to minimize the average cost?

Let $f(x) = AC = \dfrac{128{,}000 + 30x + x^3}{x}$

Step 1 : Find $f'(x)$.

$$f(x) = \frac{128{,}000}{x} + \frac{30x}{x} + \frac{x^3}{x}$$

$$= 128{,}000 \cdot x^{-1} + 30 + x^2$$

$$\Rightarrow f'(x) = 128{,}000 \cdot (-1) \cdot x^{-1-1} + 0 + 2x^{2-1}$$

$$= -128{,}000\, x^{-2} + 2x$$

Step 2 : Solve for $f'(x) = 0$

$$-128{,}000\, x^{-2} + 2x = 0$$

$$\Rightarrow 2x = 128{,}000 \cdot x^{-2}$$

$$\Rightarrow 2x = \frac{128{,}000}{x^2}$$

$$2x \cdot x^2 = 128{,}000$$

$$2x^3 = 128{,}000$$

$$x^3 = \frac{128{,}000}{2} = 64000$$

$$x = \sqrt[3]{64000} = 40$$

Step 3: Confirm that $f''(40) > 0$

$$f''(x) = \left(-128{,}000\, x^{-2} + 2x\right)'$$

$$= -128{,}000 \cdot (-2) \cdot x^{-2-1} + 2$$

$$= 256\,000\, x^{-3} + 2$$

$$f''(x) = \frac{256000}{x^3} + 2$$

$$\Rightarrow \quad f''(40) = \frac{256{,}000}{40^3} + 2 > 0$$

$x = 40$ would minimize the average cost.

Note: The AC is ALWAYS minimized when $AC = MC$

You try

The per-day cost function of the manufacture of portable MP3 players is given by

$$C(x) = 686,000 + 30x + x^3,$$

where x is the number of MP3 players manufactured per day. Assume that the company cannot manufacture more than 100 MP3 players per day.

a. Find the marginal Cost function.
b. Interpret the meaning of C' (80).
c. Find the actual cost of producing the 81st item.
d. Write the average cost function
e. How many MP3 players should be manufactured in order to minimize the average cost?

2. **Revenue Function and Marginal Revenue Function**

It represents the approximate additional income gained from selling the next unit.

Example

A luxury watch manufacturer has determined that the daily price p (in dollars) they can charge is related to the quantity q they produce by the following quadratic demand function:

$$p(x) = -x^2 + 120x + 2700$$

price

a. Find the revenue function and the marginal revenue function

Revenue = price * quantity

$R = (-x^2 + 120x + 2700) * x$

$\Rightarrow R = -x^3 + 120x^2 + 2700x$

Marginal Revenue $= MR = R'$

$\Rightarrow \quad MR = (-x^3 + 120x^2 + 2700x)'$

$MR = -3x^2 + 120 \cdot 2 \cdot x + 2700$

$MR = -3x^2 + 240x + 2700$

b. Interpret the meaning of MR(10).

$MR(10) = -3 \cdot 10^2 + 240 \cdot 10 + 2700$

$MR(10) = 4800$

$MR(10) \approx$ The revenue received when selling the 11th item.

c. Find the quantity to maximize the revenue function

$R = -x^3 + 120x^2 + 2700x$

<u>Step 1</u> : Find R'

$R' = MR = -3x^2 + 240x + 2700$

<u>Step 2</u> : Solve $R' = 0$

$-3x^2 + 240x + 2700 = 0$

$-3(x^2 - 80x - 900) = 0$

$\Leftrightarrow \quad x^2 - 80x - 900 = 0$

$\Leftrightarrow \quad (x - 90)(x + 10) = 0$

$x = 90 \quad , \quad x = -10$

since x is the quantity, it cannot be negative.

$x = 90$

<u>Step 3</u>: Confirm that $R''(90) < 0$

$R' = -3x^2 + 240x + 2700$

$\Rightarrow \quad R'' = (-3x^2 + 240x + 2700)'$

$R'' = -6x + 240$

$\Rightarrow \quad R''(90) = -6 * 90 + 240 = -300 < 0$

$\Rightarrow \quad R$ is maximized at $x = 90$

You try

A luxury watch manufacturer has determined that the daily price p (in dollars) they can charge is related to the quantity q they produce by the following quadratic demand function:

$$p(x) = -x^2 + 45x + 3000$$

a. Find the revenue function and the marginal revenue function
b. Interpret the meaning of MR (10).
c. Find the quantity to maximize the revenue function

3. Profit function and Marginal Profit Function

It represents the approximate change in total profit from producing and selling the next unit.

Example

A company determines that when x units of a product are produced each month, they will be sold at the price of

price $\rightarrow$ $$p = 22.2 - 1.2x$$

dollars per unit. The total cost of producing the x units will be

$$C(x) = 0.4x^2 + 3x + 40.$$ $\rightarrow$ cost

a. Find the profit function and the marginal profit function

P = Profit = Revenue − cost

= price × quantity − cost

$= (22.2 - 1.2x) \cdot x - (.4x^2 + 3x + 40)$

$= 22.2x - 1.2x^2 - .4x^2 - 3x - 40$

$= -1.6x^2 + 19.2x - 40$

$MP = P' = (-1.6x^2 + 19.2x - 40)'$

$\Rightarrow MP = -3.2x + 19.2$

b. Interpret the meaning of MP(10).

$MP(10) = -3.2 \times 10 + 19.2 = -12.8$

$MP(10) \approx$ the profit when selling the 11th item.

$\Rightarrow$ loses \$12.8 when selling the 11th item.

c. How many units should the company produce to maximize the profit?

$$P = -1.6x^2 + 19.2x - 40$$

Step 1: Find P'

$$P' = -3.2x + 19.2$$

Step 2: Solve $P' = 0$

$$-3.2x + 19.2 = 0$$

$$\Leftrightarrow \quad 3.2x = 19.2$$

$$x = \frac{19.2}{3.2} = 6$$

$$\boxed{x = 6}$$

Step 3: Confirm that $P''(6) < 0$

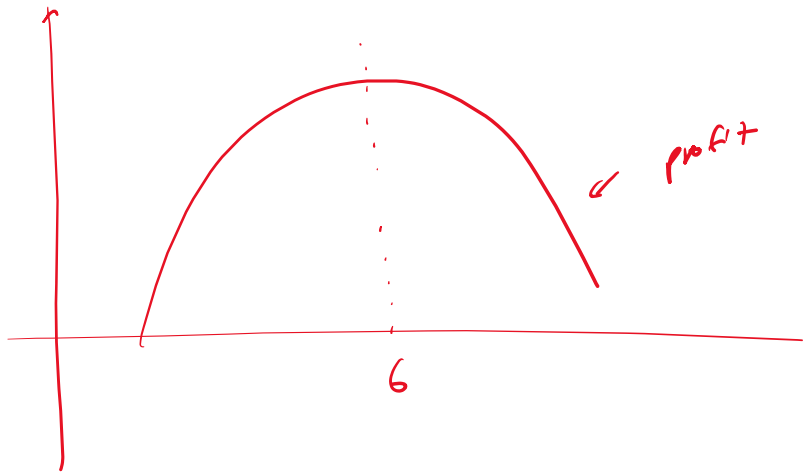
$$P' = -3.2x + 19.2$$

$$P'' = (-3.2x + 19.2)'$$

$$P'' = -3.2$$

$$\Rightarrow \quad P''(6) = -3.2 < 0$$

Profit is maximized at $x = 6$



You try

A luxury watch manufacturer has determined that the daily price p (in dollars) they can charge is related to the quantity q they produce by the following quadratic demand function:

$$p(x) = -x^2 + 45x + 3000$$

a. Find the revenue function and the marginal revenue function
b. Interpret the meaning of MR (10).
c. Find the quantity to maximize the revenue function

You try

. A company determines that when x units of a product are produced each month, they will be sold at the price of

$$p = 49 - x$$

dollars per unit. The total cost of producing the x units will be

$$C(x) = 0.125x^2 + 4x + 200.$$

a. Find the profit function and the marginal profit function
b. Interpret the meaning of MP(20).
c. How many units should the company produce to maximize the profit?